

ChessAIThon



STRATEGIC GAMING, CODING, AND AI IN VET EDUCATION

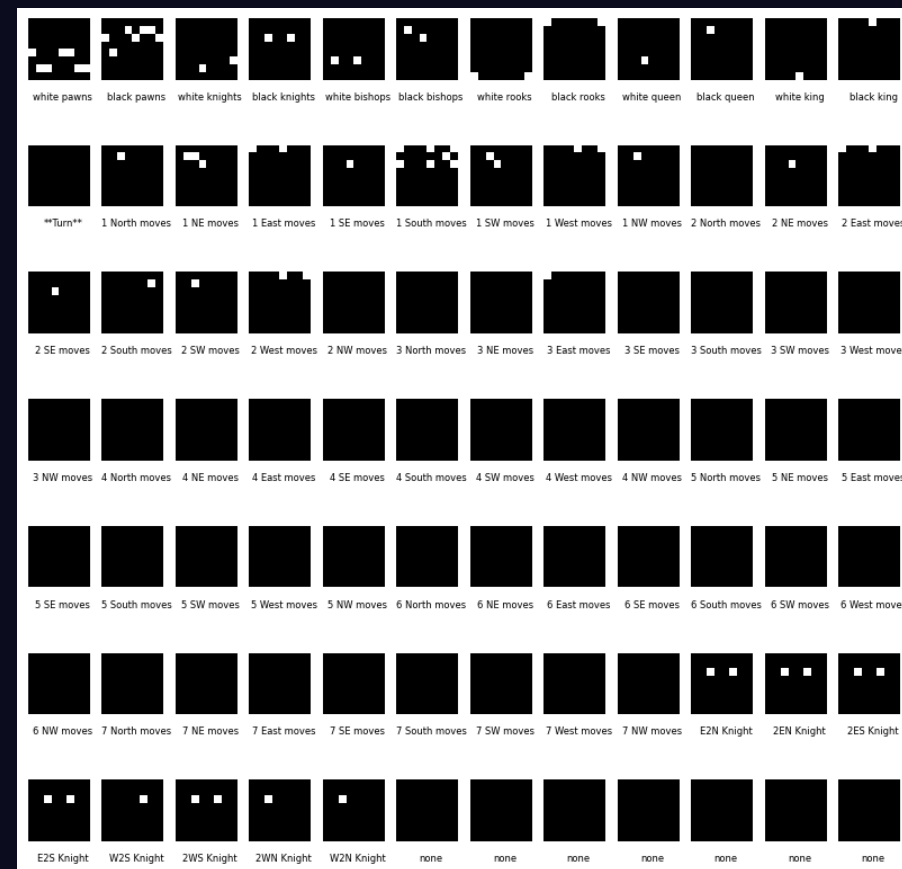
- How computers represent Chess scenarios and moves?
- How can I store scenarios and moves?
- How Chess AI works?
- How to train an AI for Chess?
- How to use this AI to get the best move?
- How can I use my intelligence to improve AI?



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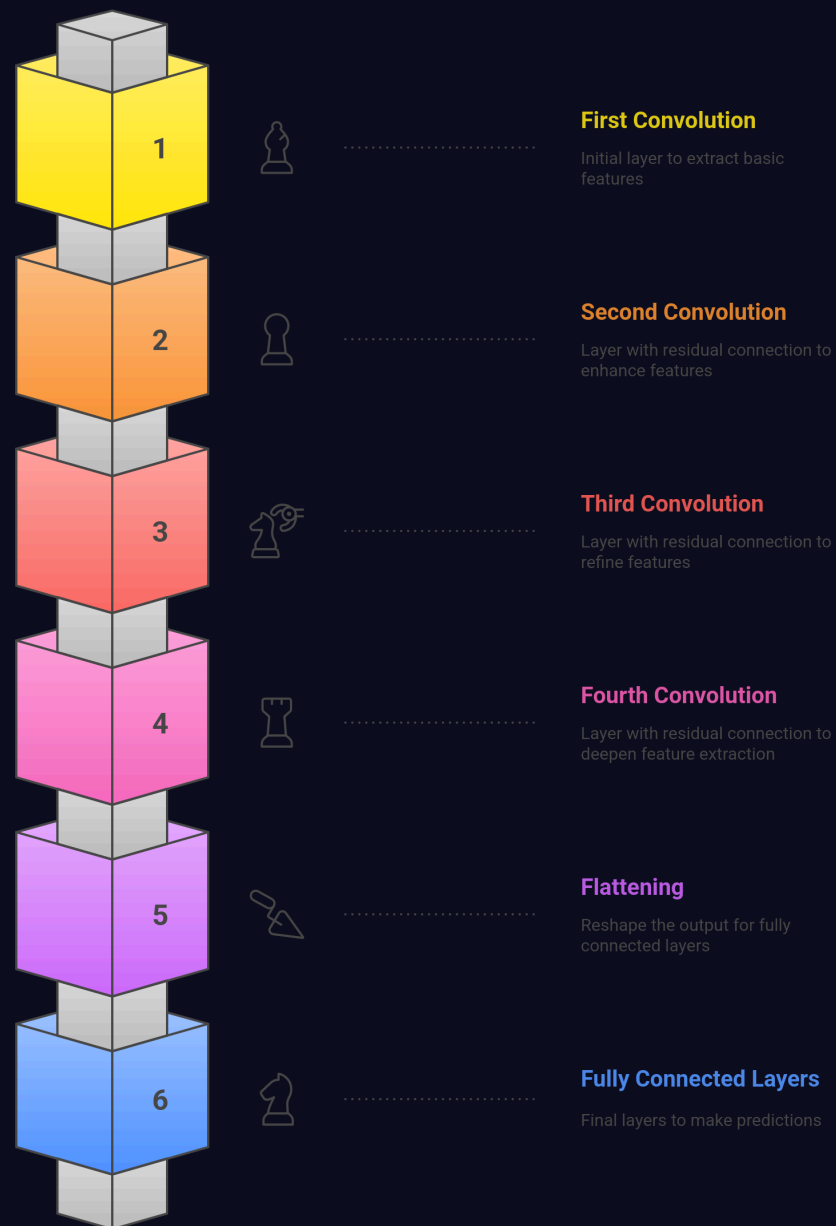
How computers represents Chess scenarios and moves?

- **Bitboards**: 64-bit integers representing squares. Highly efficient for bitwise operations.
- **Board Arrays**: Simple 8×8 grids for piece tracking.
- **Human Formats**: **FEN** (snapshot) and **UCI/SAN** (move notation).
- **AI Input (77×8×8)**:
 - 12 layers for piece positions.
 - 1 layer for turn indicator.
 - 64 layers for **legal move masks**.



How can I store scenarios and moves?

- **Standard Formats**: PGN (Games), FEN (Positions).
- **Data Science Stack**: CSV for simplicity, JSON for web APIs.
- **Parquet**: Columnar storage with **int64 compression**, reducing 10GB datasets to 1.5GB.
- **Version Control**: Using **Git/GitHub** to manage dataset iterations and collaborative student contributions.

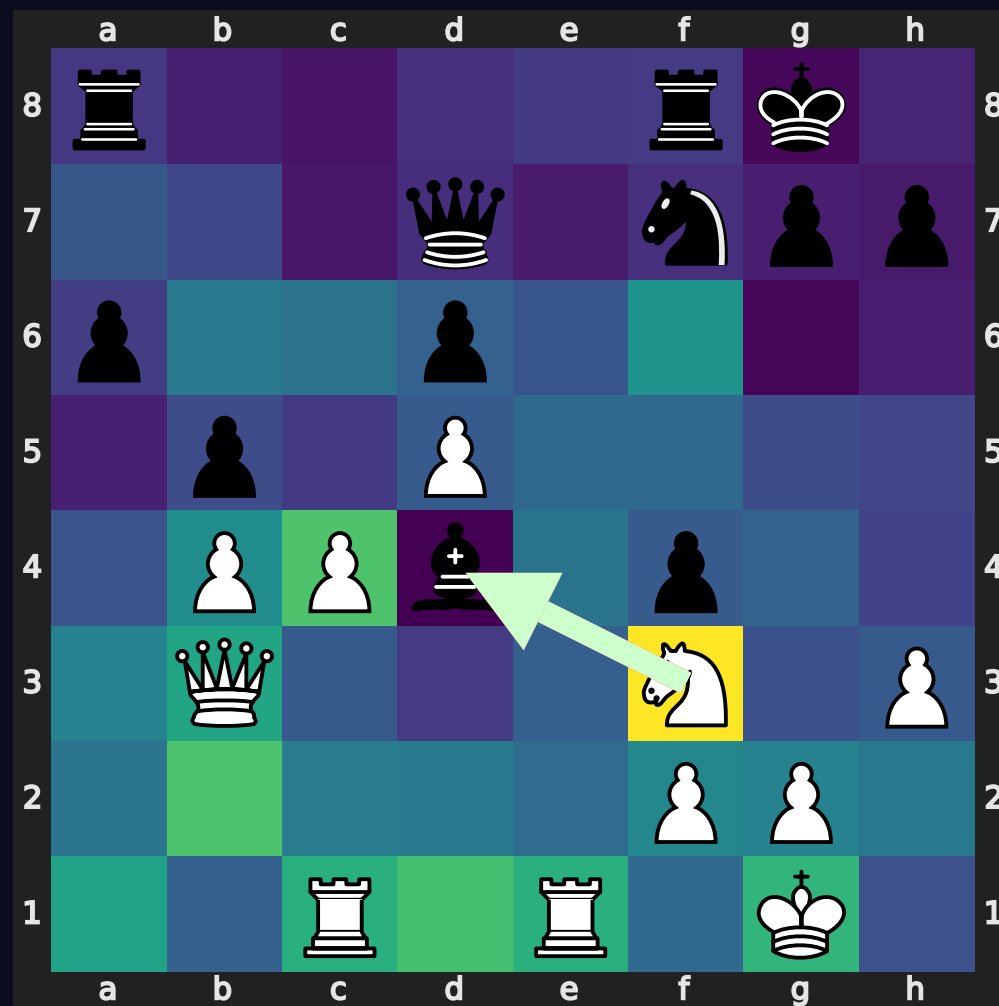


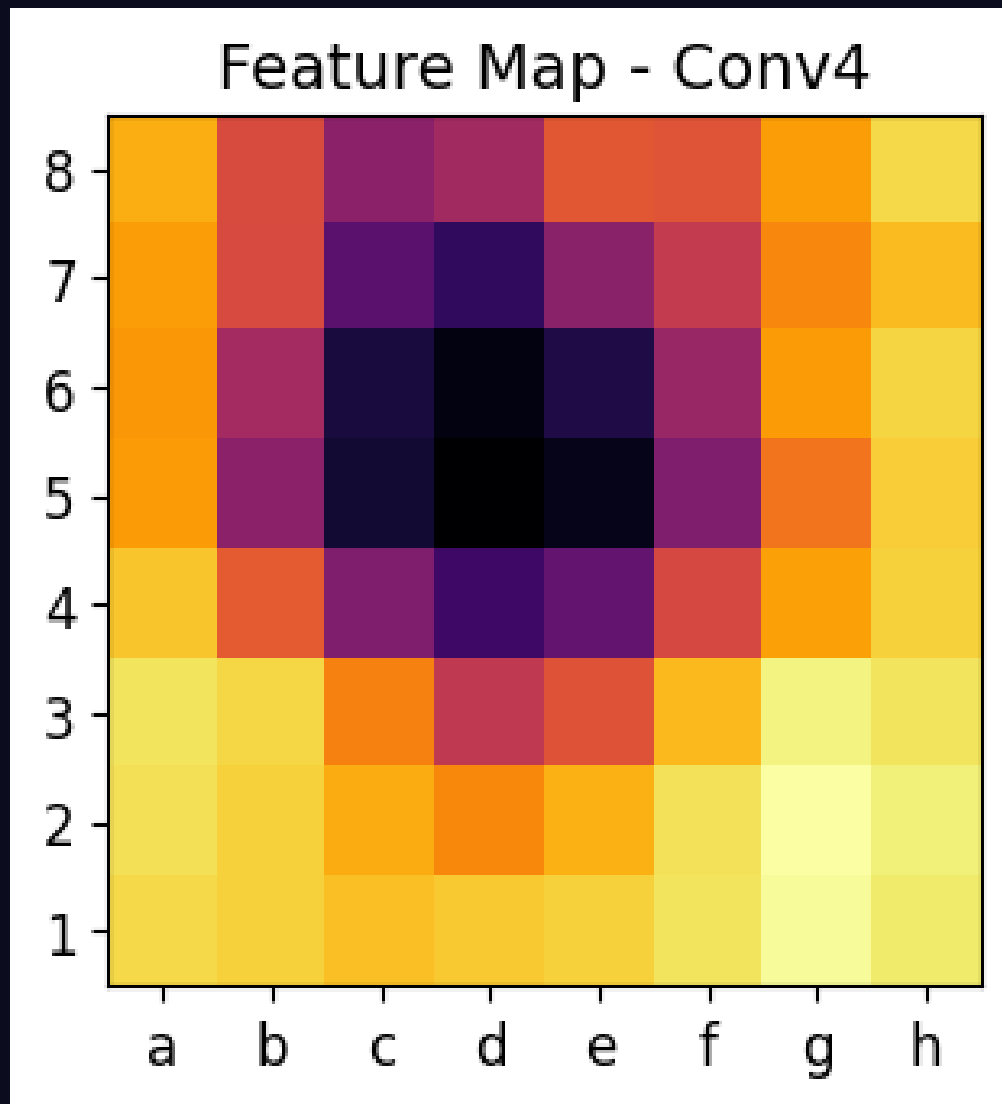
How Chess AI works?

- **Rule-based Systems:** Brute-force calculation (Deep Blue).
- **Alpha-beta Search:** Heuristic evaluation (Stockfish).
- **Modern Neural Networks:** Intuition-based move prediction (AlphaZero).
- **Hybrid Model (ChessMarro):**
 - **CNN:** Provides "intuition" (Policy + Value).
 - **MCTS:** Provides "calculation" (Look-ahead search).

How to train an AI for Chess?

- **Our Choice:** Supervised learning from a curated dataset of human and simulated games.
- **Learning from Data:** Comparing NN predictions with the "best move" targets.
- **Human-Centric:** Trained on student moves to model human-like playstyles and complexity.



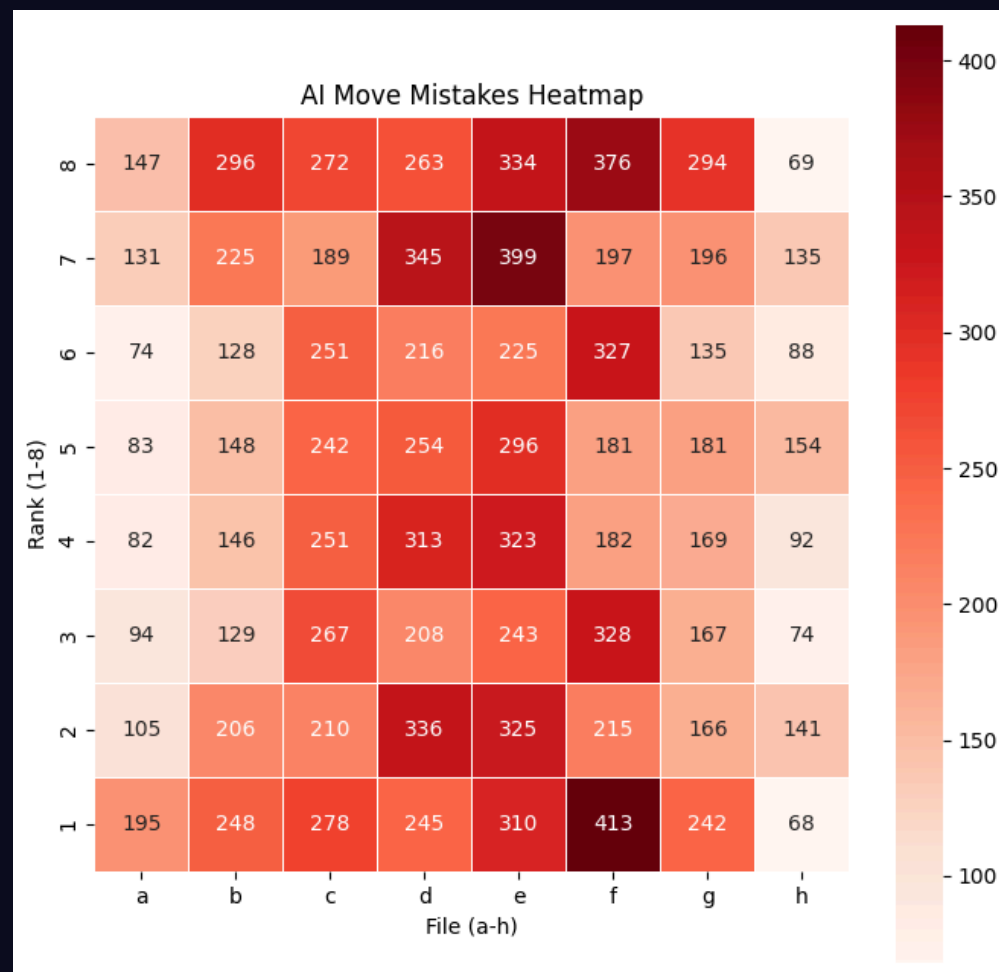


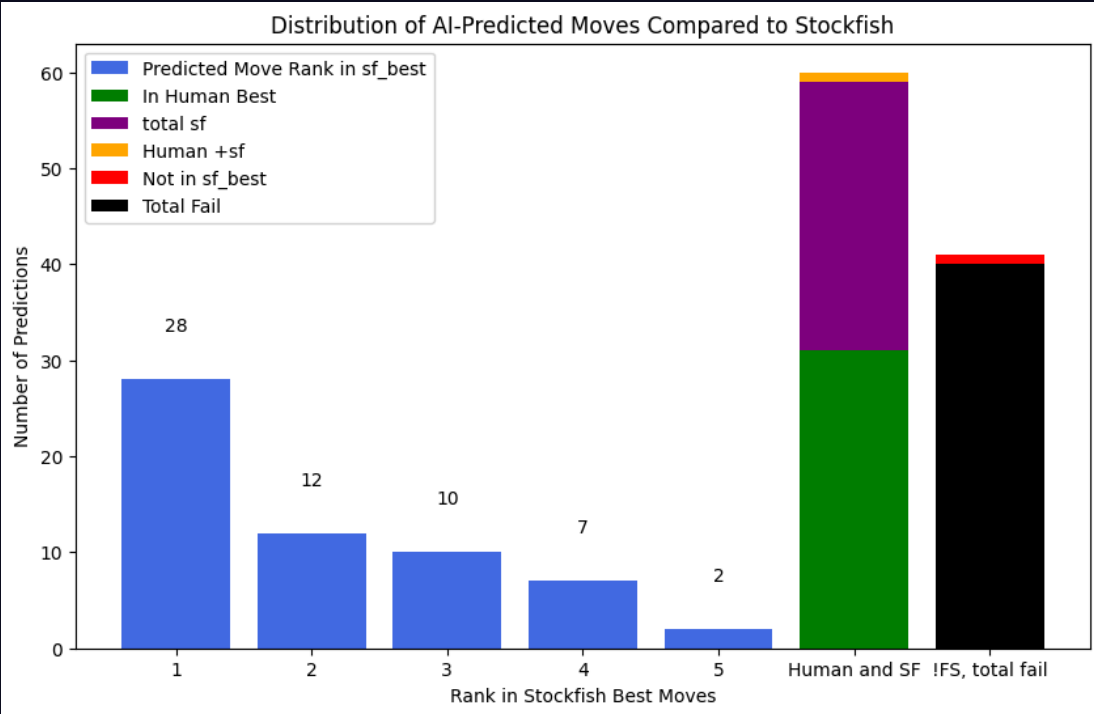
CNN

- Residual Tower
- Bitboard representation: $77 \times 8 \times 8$
- Output:
 - Policies: Probability distribution over 4096 possible moves.
 - Value: Predicting the win/loss outcome $([-1, 1])$.

Dataset

- **Source:** Leela Chess Zero simulations (400 nodes) and Lichess puzzles.
- **Volume:** ~4,500,000 scenarios in Parquet format.
- **Diversity:** Includes balanced positions, tactical puzzles, and mate-specific scenarios.
- **Refinement:** Filtering out "decided" positions to prioritize informative learning signals ($0.1 < |\text{Value}| < 0.7$).



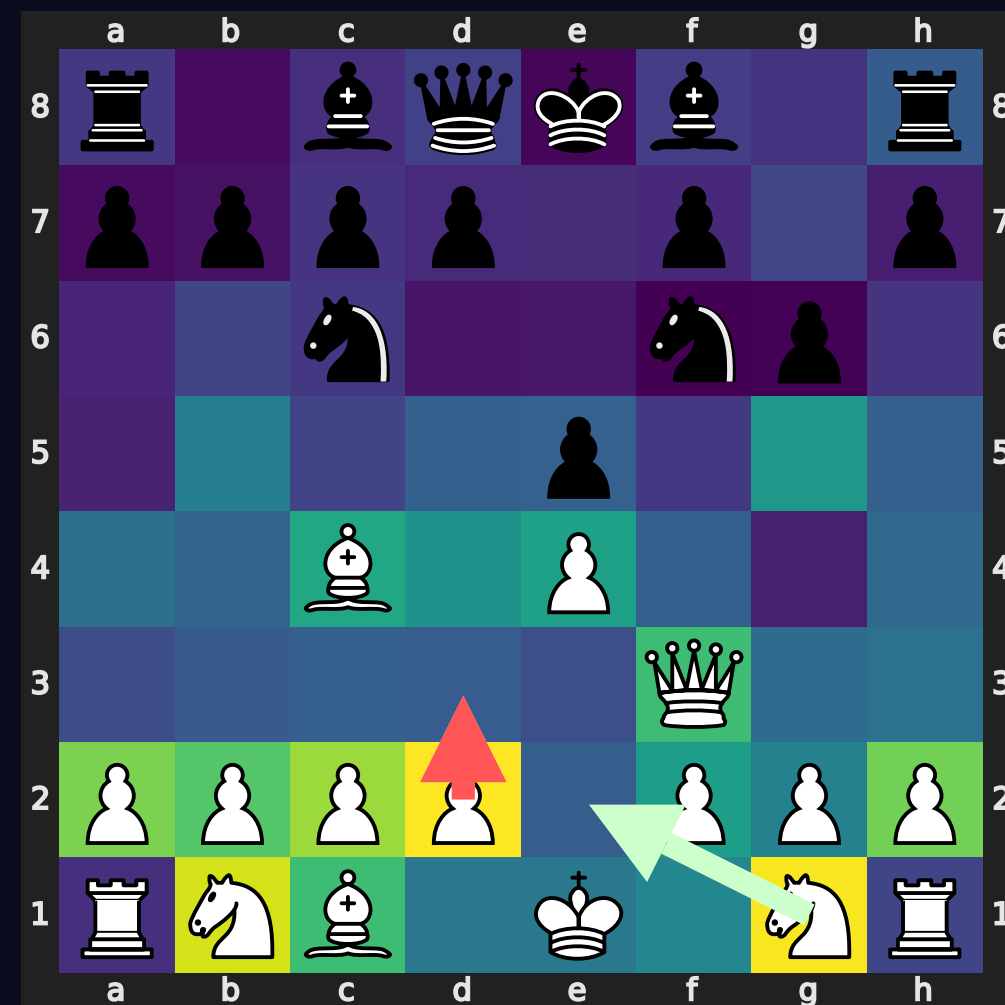


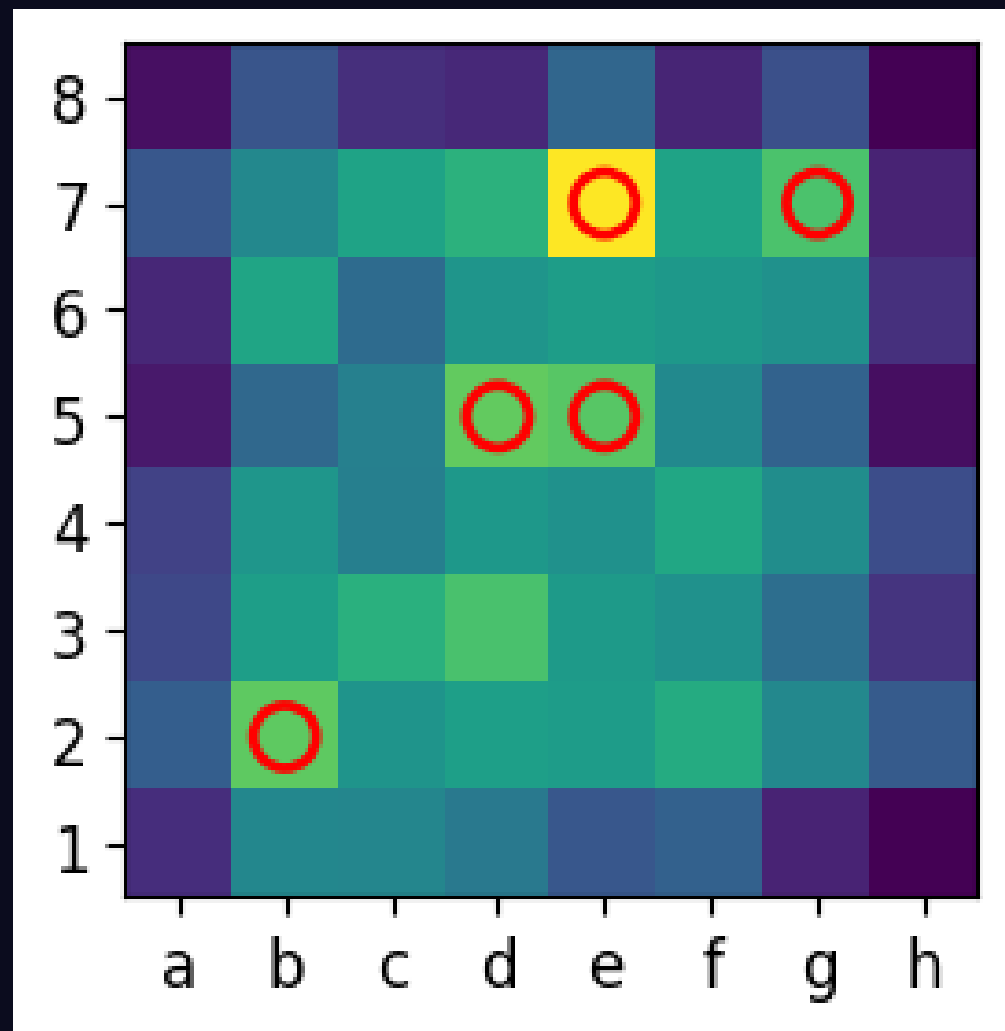
Results

- **Top-1 Accuracy: ~30%** (Matching Stockfish's perfect move).
- **Top-5 Accuracy: ~70%** (Sufficient to prune MCTS search space).
- **Human-like Behavior:** Matches human "mistakes" in complex positions, making it an ideal pedagogical tool.
- **Success:** Even modest accuracy leads to strong strategic play when paired with MCTS.

How to get the best move?

- **MCTS Guided Search**: Simulation explores branches prioritized by the CNN Policy.
- **Exploration (PUCT)**: Balancing known good moves with less-explored alternatives.
- **Noise**: Dirichlet noise at the root node ensures variety and prevents deterministic traps.
- **Mixture of experts**: Different MCTS in parallel vote the best move. The first trust in the CNN, the last more in exploration.





How can I use my intelligence to improve AI?

- **ChessMinds Web App**: Play against ChessMarro and contribute your best moves to the dataset.
- **Fine-tuning**: Use provided Notebooks to retrain the CNN on your personal games.
- **Community**: Modify architecture parameters or weights and share your version on HuggingFace.

Thank You

QUESTIONS & COLLABORATIVE AI DEVELOPMENT

GitHub: [xxjcaxx/ChessAlThon](https://github.com/xxjcaxx/ChessAlThon)